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Notifications 

16. Applications of diagonalization

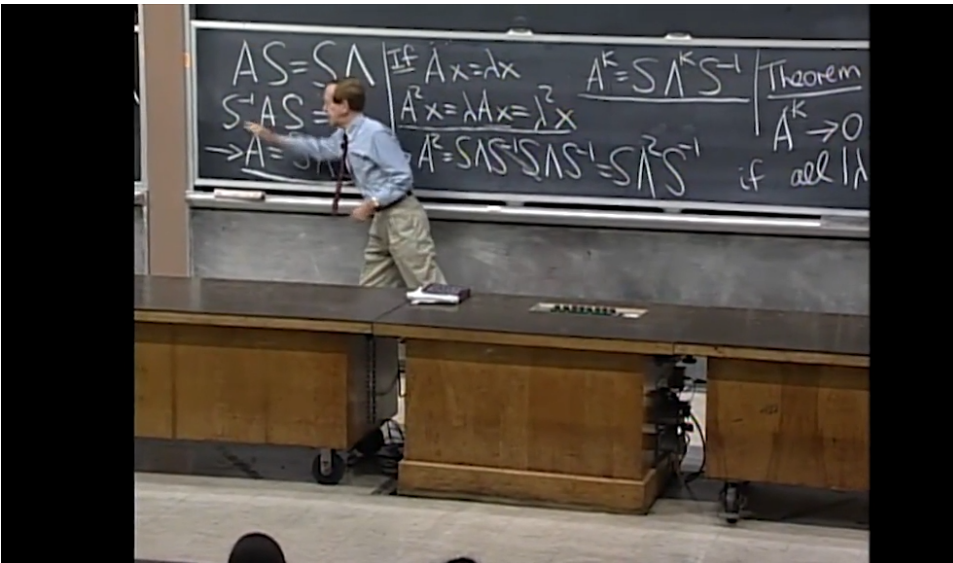
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Powers of a matrix



if I don't have that, then this approach is not working.

So again, a pure eigenvalue approach, eigenvector approach, needs n independent eigenvectors.

If we don't have n independent eigenvectors, we can't diagonalize the matrix.

We can't get to a diagonal matrix.

This diagonalization is only possible **if S inverse makes sense.**

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Suppose that $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$. Then

$$\mathbf{A}^3 = \mathbf{S}\underbrace{\mathbf{D}\mathbf{S}^{-1}\mathbf{S}}_{\text{cancels}}\underbrace{\mathbf{D}\mathbf{S}^{-1}\mathbf{S}}_{\text{cancels}}\mathbf{D}\mathbf{S}^{-1} = \mathbf{S}\mathbf{D}^3\mathbf{S}^{-1}.$$

More generally, for any integer $n \geq 0$,

$$\mathbf{A}^n = \mathbf{S}\mathbf{D}^n\mathbf{S}^{-1}. \tag{3.42}$$

Problem 16.1 Compute $\mathbf{A}^{10}$ for $\mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$.

Solution : We have previously diagonalized this matrix:

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1} \quad \text{where } \mathbf{S} = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \tag{3.43}$$

$$\mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}. \tag{3.44}$$

This gives

$$\mathbf{A}^{10} = (\mathbf{S}\mathbf{D}\mathbf{S}^{-1})^{10} \tag{3.45}$$

$$= \mathbf{S}\mathbf{D}^{10}\mathbf{S}^{-1} \tag{3.46}$$

$$= \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & (-3)^{10} & 0 \\ 0 & 0 & (-3)^{10} \end{pmatrix} \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}^{-1} \tag{3.47}$$

16. Applications of diagonalization

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